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Supplemental Table 2. Search strategies

Search strategies using MEDLINE (Supplemental Table 1a), Embase (Supplemental Table 1b), and Cochrane Cochrane Central Register of Controlled Trial (Supplemental Table 1c) are shown below.

Supplemental Table 2a. Search strategy using MEDLINE

No.	Query	Hit retrieved
S5	(('isolated noncompaction of the ventricular myocardium'/exp) OR hypertrabeculation OR LVNC OR (left ventricular trabeculation) OR (left ventricular noncompaction) OR (left ventricular non-compaction) OR (noncompacted cardiomyopathy) OR (noncompacted myocardium) OR (noncompaction cardiomyopathy) OR (noncompaction myocardium))	6,800
S6	((non-compacted cardiomyopathy) OR (non-compacted myocardium) OR (non-compaction cardiomyopathy) OR (non-compaction myocardium) OR (noncompaction of left ventricular myocardium) OR (spongy myocardium))	2,770
S8	(warfarin OR anticoagulants OR aspirin OR apixaban OR edoxaban OR rivaroxaban OR (antithrombotic therapy) OR dabigatran)	433,156
S9	(thromboembolism OR thrombosis OR embolism)	1,034,463
S10	S6 OR S5	7227

S11	S10 AND S8	219
S12	S10 AND S9	738
S13	S12 OR S11	833
S14	(S12 OR S11) and (subt.exact("human"))	744

Supplemental Table 2b. Search strategy using Embase

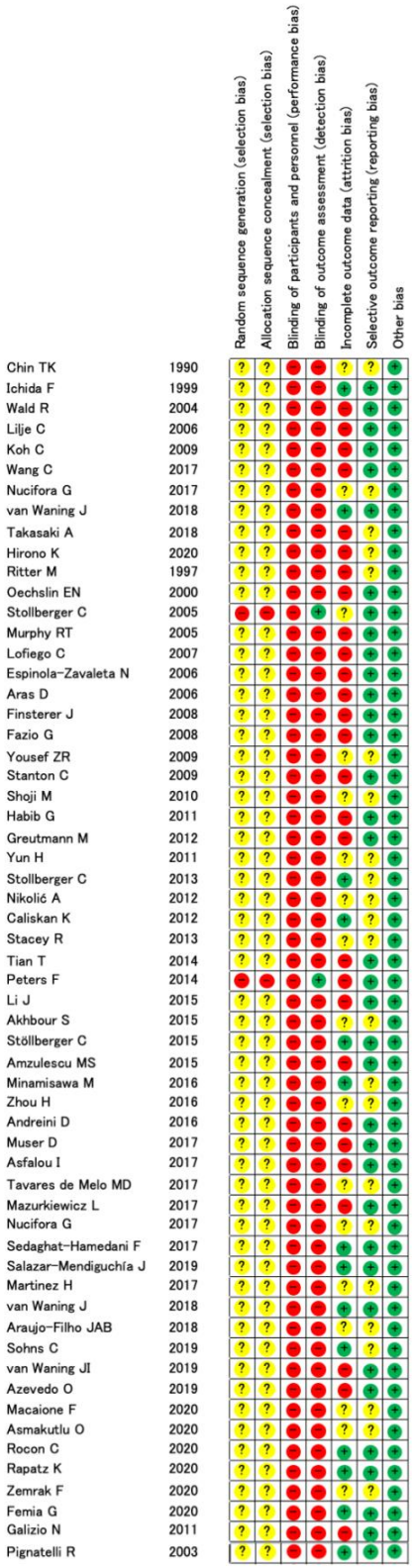
No.	Query	Hit retrieved
S1	(('isolated noncompaction of the ventricular myocardium'/exp) OR hypertrabeculation OR LVNC OR (left ventricular trabeculation) OR (left ventricular noncompaction) OR (left ventricular non-compaction) OR (noncompacted cardiomyopathy) OR (noncompacted myocardium) OR (noncompaction cardiomyopathy) OR (noncompaction myocardium)) and (human(yes))	3,754
S3	((non-compacted cardiomyopathy) OR (non-compacted myocardium) OR (non-compaction cardiomyopathy) OR (non-compaction myocardium) OR (noncompaction of left ventricular myocardium) OR (spongy myocardium)) and (human(yes))	1,842
S4	('warfarin'/exp OR 'anticoagulants'/exp OR 'aspirin'/exp OR apixaban OR edoxaban OR rivaroxaban OR (antithrombotic therapy) OR 'dabigatran'/exp) and (human(yes))	36,963
S5	(warfarin OR anticoagulants OR aspirin OR apixaban OR edoxaban OR rivaroxaban OR (antithrombotic therapy) OR dabigatran) and (human(yes))	208,200
S6	('thromboembolism'/exp OR 'thrombosis'/exp OR 'embolism'/exp) and (human(yes))	1
S7	(thromboembolism OR thrombosis OR embolism) and (human(yes))	509,254
S8	S3 OR S1	3,889
S9	S8 AND S5	184
S10	S8 AND S7	650
S11	S10 OR S9	731
S15	(S10 OR S9) and (rtype.exact("Article") AND subt.exact("human"))	339

Supplemental Table 2c. Search strategy using Cochrane Central Register of Controlled Trial

No.	Query	Hit retrieved
#1	MeSH descriptor: [Isolated Noncompaction of the Ventricular Myocardium] explode all trees	0
#2	hypertrabeculation	0
#3	LVNC	1
#4	left ventricular trabeculation	3
#5	left ventricular noncompaction	6
#6	left ventricular non-compaction	5
#7	noncompacted cardiomyopathy	2
#8	noncompacted myocardium	1
#9	noncompaction cardiomyopathy	7
#10	noncompaction myocardium	4

#11	non-compacted cardiomyopathy	2
#12	non-compacted myocardium	1
#13	non-compaction cardiomyopathy	6
#14	non-compaction myocardium	4
#15	noncompaction of left ventricular myocardium	4
#16	spongy myocardium	0
#17	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR 15 OR 16	448,627
#18	MeSH descriptor: [Warfarin] explode all trees	1,700
#19	MeSH descriptor: [Anticoagulants] explode all trees	4,663
#20	MeSH descriptor: [Aspirin] explode all trees	5,935
#21	apixaban	978
#22	MeSH descriptor: [Dabigatran] explode all trees	312
#23	edoxaban	594
#24	rivaroxaban	1,738
#25	antithrombotic therapy	2,119
#26	#18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25	14,025
#27	MeSH descriptor: [Thromboembolism] explode all trees	2,112
#28	MeSH descriptor: [Thrombosis] explode all trees	4,841
#29	MeSH descriptor: [Embolism] explode all trees	1,427
#30	#27 OR #28 OR #29	7,272
#31	#17 AND #26	1
#32	#17 AND #30	0
#33	#31 OR #32	1

Supplemental Table 3. Risk of bias table



According to the Cochrane Hand book for Systematic Reviews of Interventions, version 5.1.0. +: low risk of bias; - : high risk of bias; ?: unclear risk of bias.

Supplemental Table 4. Characteristics of included studies

Author	Year	Patients (n)	Age (yeras)	Male (%)	Follow-up period (years)	Diagnostic criteria	Echo/MRI	LVEF (%)	AF (%)	VT (%)	Anticoagulant	Antiplatelet	Anticoagulant/ Antiplatelet (%)	Hx of TE (%)	TE events (%)
Children															
Chin[1]	1990	8	8.9 (0.9−22.5)	63.0	N/A	Chin	Echo	N/A	0	62.5	N/A	N/A	N/A	38.0	N/A
Ichida[2]	1999	27	5.0 (0−15.0)	55.6	6.0 (0−17.0)	Ichida	Echo	61.5 ± 11.6	3.7	0	N/A	N/A	N/A	N/A	0
Pignatelli[3]	2003	36	0.3 (0-17)	55.6	3.2 (0.5- 12.0)	Jenni	Echo	25 (18.0-32.0)	0	2.8	N/A	N/A	N/A	2.8	0
Wald[4]	2004	22	3.9 (0−16.0)	40.9	3.0 (0.1−16)	Jenni	Echo	34.0 (6.0−67)	0	9.1	N/A	N/A	N/A	N/A	0
Lilje[5]	2006	66	4.0 (0−21.0)	48.5	1.0 (0−4.3)	Chin, Jenni	Echo	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13.9
Koh[6]	2009	10	2.0 (0−12.0)	70.0	2.0 (0.2−3.0)	Jenni	Echo	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0
Wang[7]	2017	205	3.6 (0−15.0)	53.7	4.6 (0−22.0)	Jenni	Echo	49.9 ± 2.1	N/A	N/A	N/A	N/A	N/A	N/A	4.9
Nucifora[8]	2017	12	15.0 ± 3.0	75.0	N/A	Petersen, Jacquire	MRI	64.0 ± 8.0	N/A	8.0	N/A	N/A	N/A	0	N/A
van Waning[9]	2018	52	7.0 (0−14.0)	52.0	5.0 (1.5−9.4)	Jenni, Petersen	UCG/MRI	N/A	10.0	6.0	N/A	N/A	N/A	2.0	4.0
Takasaki[10]	2018	82	0.8 (0−35.0)	56.1	4.8 (0−22.0)	Paterick	Echo	42.4 ± 3.5	N/A	N/A	N/A	N/A	N/A	3.7	N/A
Hirono[11]	2020	206	0.3 (0−16.0)	57.8	2.0 (0−33.0)	Paterick	Echo	47.1 ± 19.1	0	11.2	N/A	N/A	N/A	2.4	N/A
Adults															
Ritter[12]	1997	17	41.6 (18.0−71.0)	82.3	2.5 ± 2.3	Stollberger	Echo	N/A	29.4	11.8	N/A	N/A	N/A	5.9	24.0
Oechslin[13]	2000	34	42.0 ± 17.0	74.0	3.7 ± 3.3	Jenni	Echo	33.0 ± 13.0	26.0	41.0	N/A	N/A	N/A	N/A	24.0
Stollberger[14]	2005	62	53.0	86.0	N/A	Stollberger	Echo	FS 25.1	8.1	N/A	yes	N/A	6.5	1.6	10.0
Murphy[15]	2005	45	37.0 ± 17.0	62.2	2.7 (0.5−14.9)	Chin, Jenni	Echo	FS 21.0 ± 9.0	6.5	22.0	yes	N/A	60.0	N/A	4.0
Lofiego[16]	2007	65	42.0 ± 17.0	N/A	3.8 (0.5−16.1)	Jenni	Echo	31.0 ± 11.0	1.5	6.0	yes	N/A	52.3	N/A	5.0
Espinola-Zavaleta[17]	2006	53	43.2 ± 14.8	47.2	0.6 ± 0.4	Jenni	Echo	39.0 ± 18.5	5.7	7.5	N/A	N/A	N/A	N/A	5.7
Aras[18]	2006	67	41.0 ± 18.0	66.0	3.0 ± 1.1	Jenni	Echo	43.5 ± 14.4	12.0	36.0	N/A	yes	28.3	6.0	9.0
Finsterer[19]	2008	104	14.0−94.0	71.0	12.0	Stollberger	Echo	N/A	N/A	N/A	N/A	N/A	N/A	9.0	15.0
Fazio[20]	2008	238	41.5 (1.0−92.0)	58.0	7.3 (1.0−12.0)	Stollberger	Echo	24.6	N/A	N/A	N/A	N/A	N/A	14.0	2.1
Yousef[21]	2009	42	48.7 ± 2.3	59.5	N/A	Jenni	Echo	41.1 ± 14.8	28.6	N/A	N/A	N/A	N/A	14.0	N/A
Stanton[22]	2009	30	39.0 ± 19.5	60.0	2.5 ± 1.2	Jenni	Echo	41.0 ± 15.0	16.6	10.0	N/A	yes	30.0	0	0
Shoji[23]	2010	187	41.3 ± 16.8	65.8	N/A	Jenni	Echo	62.2	4.8	1.1	N/A	N/A	N/A	0.5	N/A
Habib[24]	2011	105	45.0 ± 17.0	66.0	2.3 ± 1.5	Jenni	Echo	46.0 ± 18.0	7.0	7.0	yes	N/A	29.2	4.0	8.5
Greutmann[25]	2012	132	41.0 ± 17.0	35.0	2.7 (0.1−19.4)	Jenni	Echo	41.0 ± 18.0	24.0	7.0	yes	yes	36.0	3.0	4.0
Yun[26]	2011	11	35.0 (15.0−58.0)	81.8	N/A	Petersen	MRI	40.0 (23.0−70.0)	0	0	N/A	N/A	N/A	9.1	N/A
Stollberger[27]	2013	169	52.9 ± 16.0	70.0	16.0	Stollberger	Echo	FS 24.2 ± 11.0	16.6	N/A	N/A	N/A	N/A	15.0	N/A
Nikolić[28]	2012	12	45.0 ± 15.0	58.3	N/A	Jenni	Echo	49.8 ± 8.0	33.3	25.0	N/A	N/A	N/A	0	N/A
Caliskan[29]	2012	84	40.0 (17.0−77.0)	47.6	5.3 (1.5−9.6)	Jenni	Echo	FS 20.8 ± 8.2	9.5	19.0	N/A	N/A	N/A	10.7	N/A
Stacey[30]	2013	122	57.0 ± 17.5	59.0	N/A	Jacquier	MRI	44.0 ± 16.0	N/A	4.9	N/A	N/A	N/A	8.2	N/A
Tian[31]	2014	106	46.0 ± 17.0	78.0	2.9 ± 2.1	Jenni	Echo	39.0 ± 14.0	13.0	2.0	yes	N/A	9.0	6.0	4.0
Peters[32]	2014	55	42.2 ± 11.5	38.2	1.4 ± 0.5	Jenni	Echo	29.6 ± 11.8	1.8	1.8	yes	N/A	29.1	16.3	1.8

Author	Year	Patients (n)	Age (yeras)	Male (%)	Follow-up period (years)	Diagnostic criteria	Echo/MRI	LVEF (%)	AF (%)	VT (%)	Anticoagulant	Antiplatelet	Anticoagulant/ Antiplatelet (%)	Hx of TE (%)	TE events (%)
Li[33]	2015	20	42.5 (36.3−52.5)	60.0	1.1±0.5	Stollberger	Echo	32.0 (22.0−40.0)	5.0	5.0	yes	yes	60.0	N/A	0
Akhbour[34]	2015	24	42.7 ± 13.1	66.7	N/A	Petersen	MRI	37.8 ± 13.9	16.7	20.8	N/A	N/A	N/A	4.2	N/A
Stöllberger[35]	2015	232	52.0 ± 17.0	69.0	6.0 ± 5.2	Stollberger	Echo	FS 25.0 ± 11.0	16.0	N/A	N/A	N/A	N/A	N/A	4.3
Amzulescu[36]	2015	59	52.0 ± 13.0	57.6	3.4 (1.5 − 6.3)	Petersen	MRI	24.1 ± 8.3	3.0	N/A	yes	yes	22.0	12.0	6.8
Minamisawa[37]	2016	23	54.9 ± 9.1	69.6	5.1	Jenni	Echo	27.4 ± 8.5	43.5	N/A	yes	N/A	43.4	8.7	N/A
Zhou[38]	2016	31	40.8 ± 4.8	61.3	N/A	Jenni	Echo	42.0 ± 5.9	N/A	N/A	N/A	N/A	N/A	6.5	N/A
Andreini[39]	2016	113	44.0 ± 17.0	62.0	4 ± 2	Jenni, Petersen	Echo /MRI	42.8 ± 16.2	N/A	N/A	yes	yes	5.0	N/A	4.4
Muser[40]	2017	9	42.0 ± 15.0	55.6	4.0 (1.0 − 11.0)	Jenni	Echo	40.0 ± 12.1	N/A	44.4	yes	N/A	66.6	N/A	11.1
Asfalou[41]	2017	23	47.0 ± 13.0	65.2	2.0	Jenni	Echo	27.0 ± 8.0	21.7	16.6	yes	N/A	28.6	N/A	4.3
Tavares de Melo[42]	2017	30	41.0 ± 12.0	53.0	N/A	Petersen	MRI	39.0 ± 14.0	N/A	33.0	yes	N/A	77.0	13.0	N/A
Mazurkiewicz[43]	2017	127	33.1 ± 8.8	61.4	2.2 ± 1.0	Grothoff	MRI	27.7 ± 6.8	11.8	N/A	N/A	N/A	N/A	8.7	1.6
Nucifora[8]	2017	20	35.0 ± 7.0	75.0	N/A	Petersen, Jacquire	MRI	55.0 ± 14.0	N/A	10.0	N/A	N/A	N/A	10.0	N/A
Sedaghat-Hamedani[44]	2017	95	41.0 ± 14.4	70.6	5.1	Jenni, Stollberger	Echo	38.0 ± 15.3	29.4	35.3	N/A	N/A	N/A	N/A	10.3
Salazar-Mendiguchía[45]	2019	75	50.4 ± 14.8	68.0	5.0 (2.4 − 6.7)	Jenni	Echo	32.0 (29.0 − 34.0)	38.7	4.0	yes	yes	53.3	1.3	5.3
Martinez[46]	2017	13	N/A	N/A	N/A	Jenni	Echo	N/A	N/A	N/A	N/A	yes	N/A	8.0	N/A
van Waning[9]	2018	275	45.0 (33.0 − 56.0)	54.0	2.1 (0.3 − 4.8)	Jenni, Petersen	Echo /MRI	N/A	16.0	6.0	N/A	N/A	N/A	3.0	13.0
Araujo-Filho[47]	2018	36	41.0 ± 13.0	61.0	N/A	Petersen	MRI	42.0 ± 6.0	N/A	30.6	yes	N/A	55.6	8.3	N/A
Sohns[48]	2019	18	43.5 ± 18.0	66.7	5.2 ± 3.5	Jenni	Echo	35.0 ± 15.0	11.1	66.7	yes	N/A	77.8	27.8	N/A
van Waning[49]	2019	216	38.0 (23.0 − 52.0)	54.0	3.7 (0.8 − 7.8)	Jenni, Petersen, Grothoff	Echo /MRI	N/A	11.6	11.6	N/A	N/A	N/A	N/A	4.6
Azevedo[50]	2019	78	47.0 ± 17.0	62.8	4.4 ± 2.8	Jenni, Stollberger, Chin, Petersen Jacquier	Echo /MRI	47.2 ± 15.2	13.5	21.6	N/A	N/A	N/A	5.1	11.5
Macaione[51]	2020	27	47.6 ± 22.6	77.8	N/A	Jenni, Petersen, Grothoff	Echo /MRI	48.5 ± 17.6	N/A	11.1	N/A	N/A	N/A	3.7	N/A
Asmakutlu[52]	2020	42	32.4 (18.0 − 63.0)	52.3	N/A	Stollberger, Petersen	Echo /MRI	50.3 (22.0 − 67.0)	4.7	11.9	N/A	N/A	N/A	14.2	N/A
Rocon[53]	2020	108	38.3 ± 15.5	48.1	5.8 ± 3.9	Chin, Jenni, Stollberger, Petersen	Echo /MRI	45.3 ± 14.6	N/A	61.7	N/A	N/A	N/A	N/A	17.0
Rapatz[54]	2020	113	57.0 (45.0 − 69.0)	76.0	6.1 ± 5.3	Stollberger	Echo	FS 20.0 (15.0 − 30.0)	20.0	N/A	N/A	N/A	N/A	12.0	6.0
Zemrak[55]	2020	142	N/A	N/A	N/A	Petersen	MRI	N/A	N/A	N/A	N/A	N/A	N/A	1.3	N/A
Femia[56]	2020	98	47.0 (31.0 − 65.0)	56.7	6.5 (5.5 − 7.5)	Petersen	MRI	81.2 ± 7.2	1.0	16.3	N/A	N/A	N/A	7.1	4.1
Galizio[57]	2011	80	41.0 ± 17.0	66.3	1.1 (0.5 − 2.1)	Chin, Jenni, Stollberger, Petersen	Echo /MRI	38.6 ± 11.4	N/A	18.8	N/A	N/A	N/A	N/A	3.75

Echo, echocardiography; MRI magnetic resonance imaging; LVEF, left ventricular ejection fraction; AF; atrial fibrillation; VT, ventricular tachycardia; Hx, history; TE, thromboembolism; N/A, not applicable; FS, fractional shortening.

Supplemental Table 5. Result of multiple linear regression analysis coefficients for logarithmic-transformed prevalence of thromboembolism

	Unstandardized coefficients				Standardized coefficients			
Model	β	Standard error	Lower CI	Upper CI	β	T	P value	Variance Inflation Factor
Constant	3.0518441	0.672898	1.5296435	4.5740446	0	4.54	0.0014	
Follow-up period (y)	0.386106	0.164966	0.0129269	0.7592851	0.730529	2.34	0.0440	1.8558142
LVEF (%)	-0.046194	0.017301	-0.085332	-0.007057	-0.83592	-2.67	0.0256	1.8671364
AF (%)	-0.053726	0.018199	-0.094894	-0.012558	-0.90918	-2.95	0.0162	1.8067165

CI, confidence interval; LVEF, left ventricular ejection fraction; AF; atrial fibrillation.

Supplemental Table 6. Comparison of clinical characteristics between pediatric patients with and without TE

	TE negative (n = 281)	TE positive (n = 6)	*P* value
Male	142	4	1.0000
FH	72	2	1.0000
CHD	37	0	1.0000
HF at diagnosis	154	5	0.2303
LVEF (%)	46.9 ± 18.9	30.3 ± 19.9	0.0648
LVEF<40%	89	4	0.0608
AF	2	0	1.0000
VT	29	1	0.4876
Anticoagulant	3	1	0.0298
Antiplatelet	7	0	1.0000

FX, family history; CHD, congenital heart disease; HF, heart failure; LVEF, left ventricular ejection fraction; AF, atrial fibrillation; VT, ventricular tachycardia

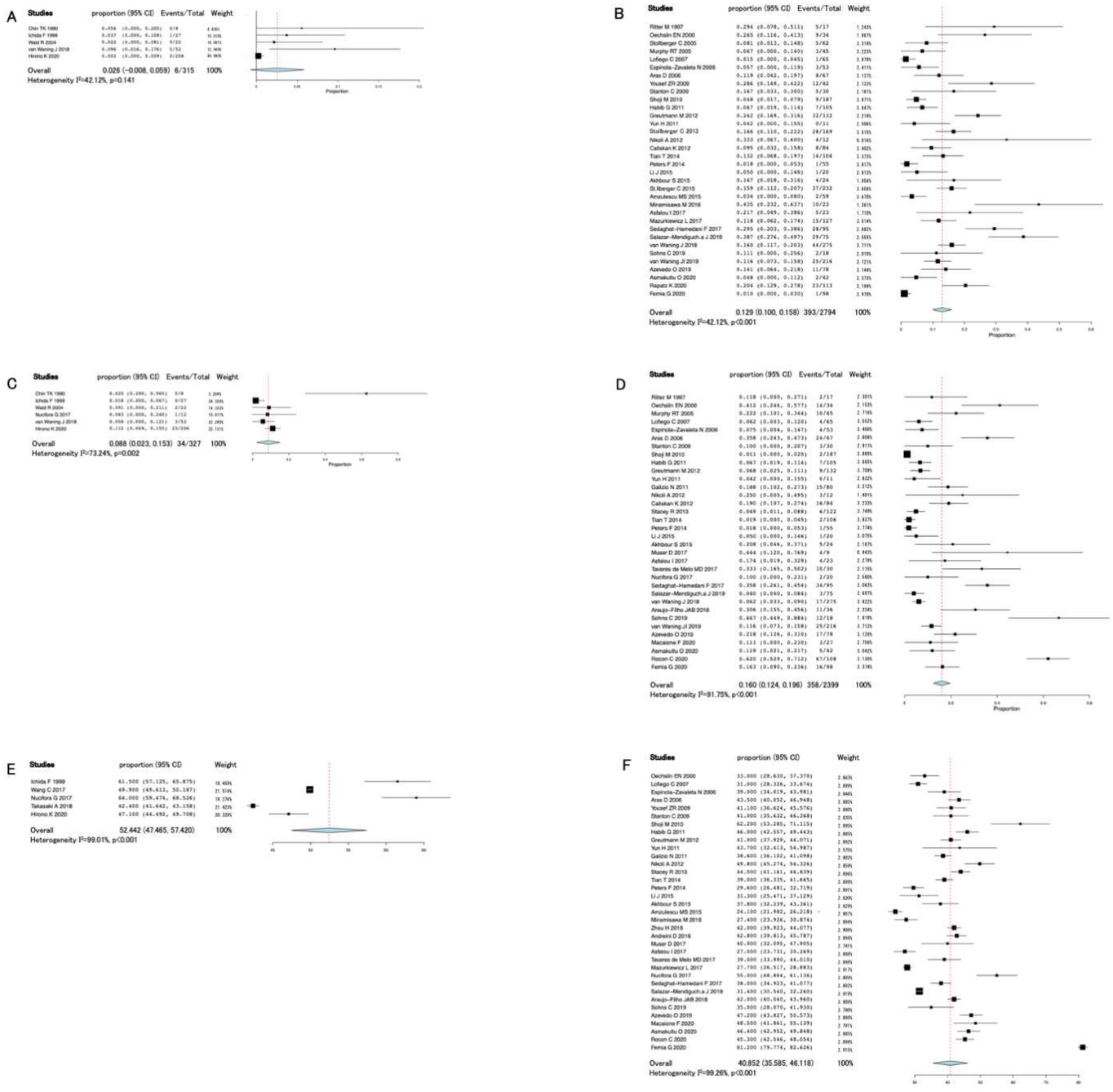
Supplemental Table 7. Comparison of clinical characteristics between adult patients with and without TE

	TE negative (n = 21)	TE positive (n = 5)	*P* value
Male	14	5	0.2782
FH	2	1	0.3333
CHD	0	0	1.0000
HF at diagnosis	7	3	0.3402
LVEF (%)	39.8 ± 13.8	40	1.0000
LVEF<40%	3	0	1.0000
AF	4	1	1.0000
VT	9	1	0.6169

FX, family history; CHD, congenital heart disease; HF, heart failure; LVEF, left ventricular ejection fraction; AF, atrial fibrillation; VT, ventricular tachycardia

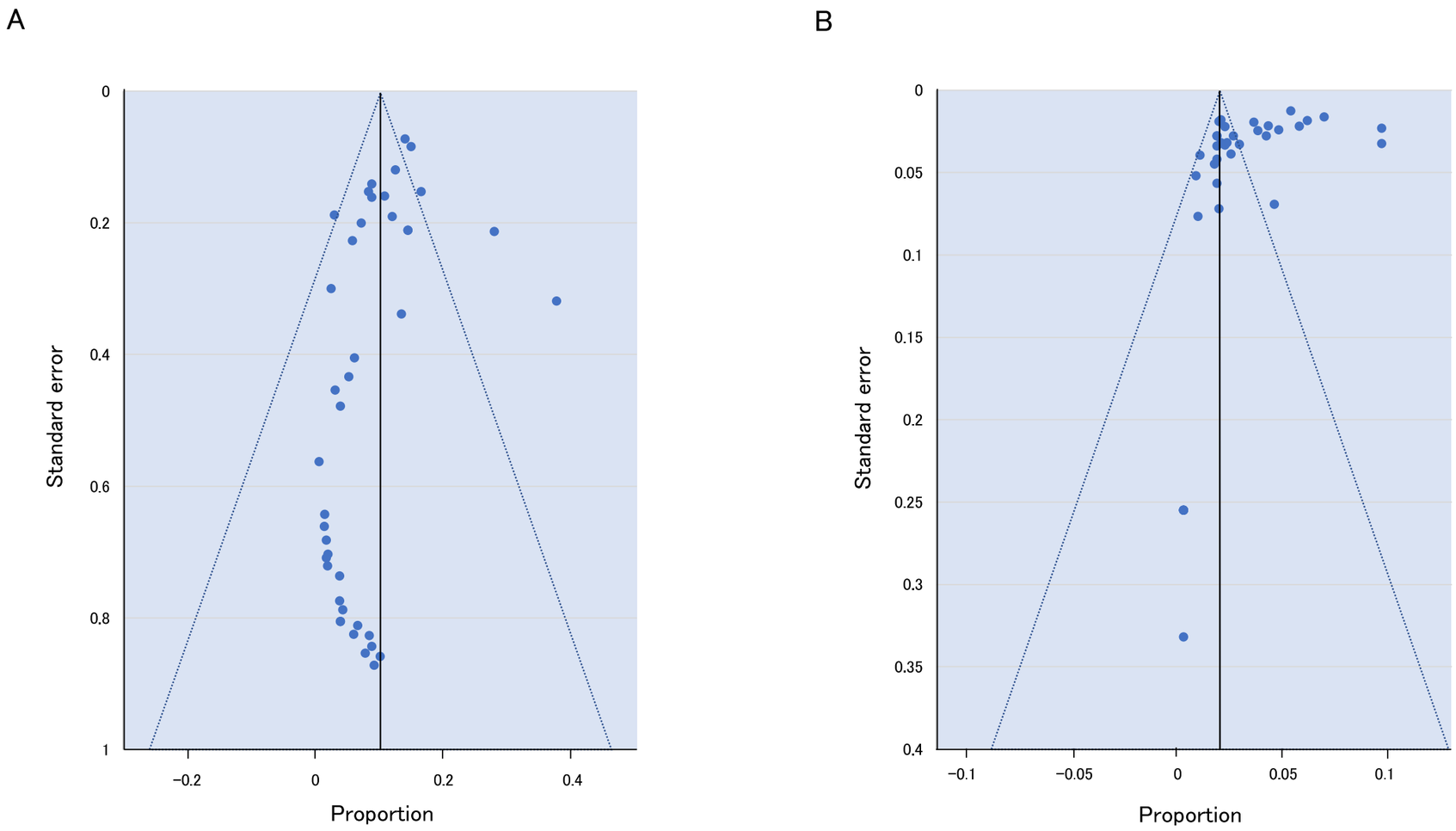
Supplemental Figure 1. Summaries of forest plots of the mean prevalence of arrhythmia and LVEF in patients with LVNC

Mean prevalence of atrial fibrillation (AF) in pediatric (A) and adult patients (B). Mean prevalence of ventricular tachycardia (VT) in pediatric (C) and adult patients (D). Mean LVEF in pediatric (E) and adult patients (F).



Supplemental Figure 2. Funnel plot of the assessment of publication bias for studies

Funnel plot of the assessment of publication bias for studies regarding the prevalence (A) and incidence (B) of TE in the patients with LVNC



Supplemental Figure 3. Forest plot of the mean prevalence of thromboembolism in the patients with LVNC according to subgroup analysis

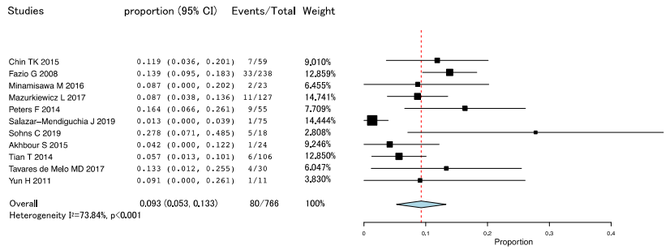
Mean prevalence of thromboembolism in studies with average LVEF <40% (A) and >40% (B).

Mean prevalence of thromboembolism in studies with mean follow-up period <4years (C) and >4years (D).

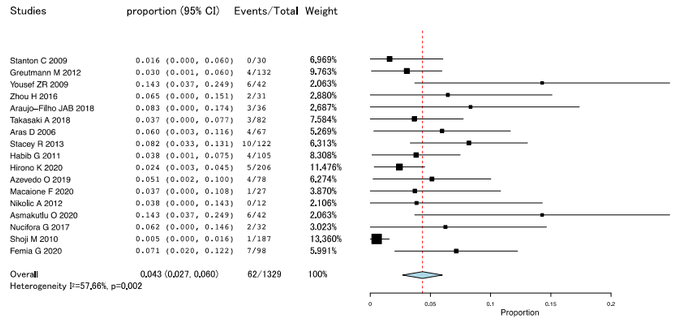
Mean prevalence of thromboembolism in studies with mean prevalence of AF <10% (E) and prevalence of AF >10% (F).

CI, confidence interval.

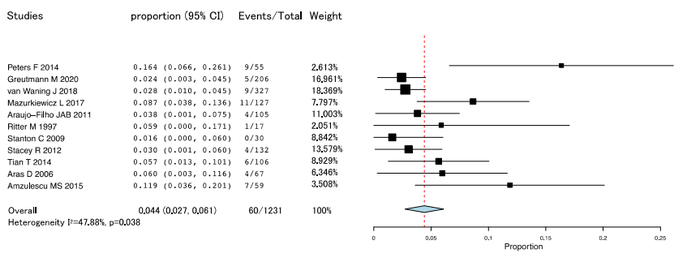
A



B



C



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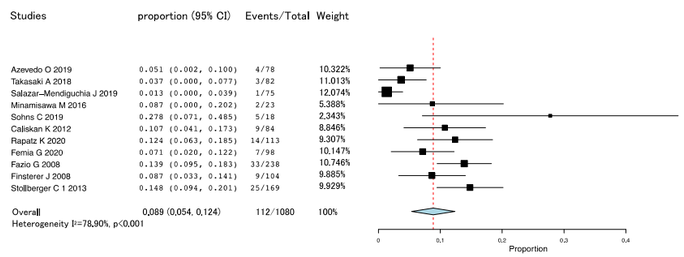

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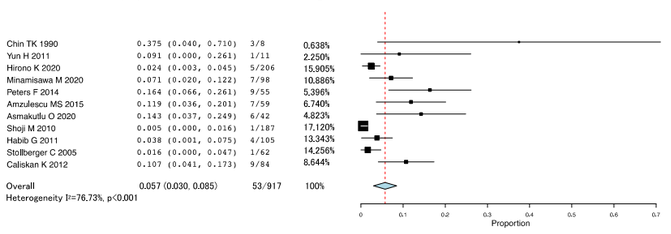

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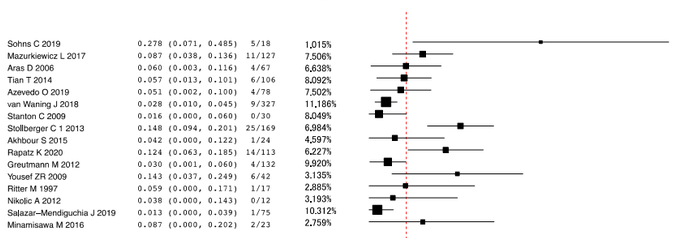


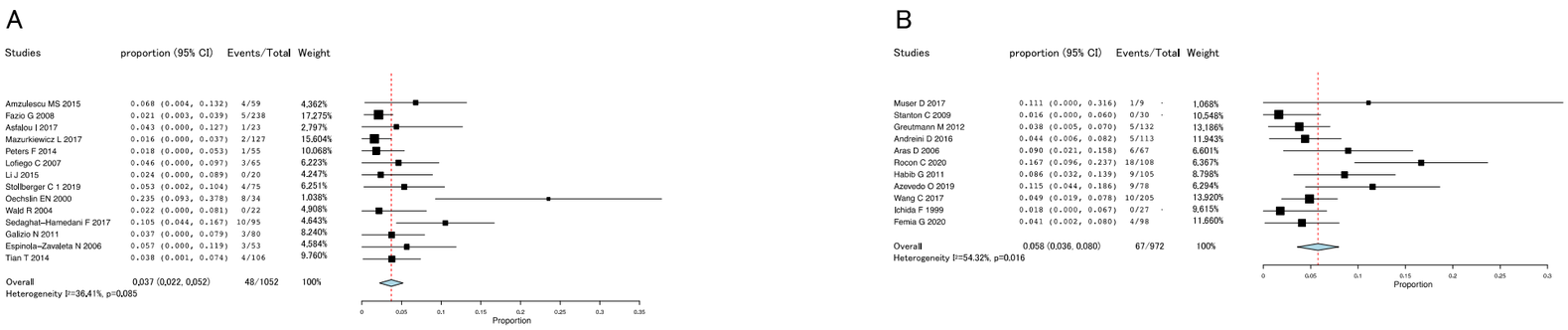
Supplemental Figure 4. Forest plot of the mean incidence of thromboembolism in the patients with LVNC according to subgroup analysis

Mean incidence of thromboembolism in studies with average LVEF <40% (A) and >40% (B).

Mean incidence of thromboembolism in studies with mean follow-up period <4years (C) and >4years (D).

Mean incidence of thromboembolism in studies with mean prevalence of AF <10% (E) and prevalence of AF >10% (F).

CI, confidence interval.

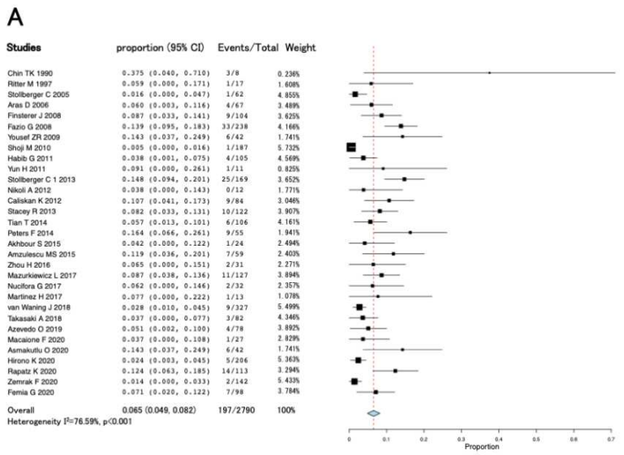


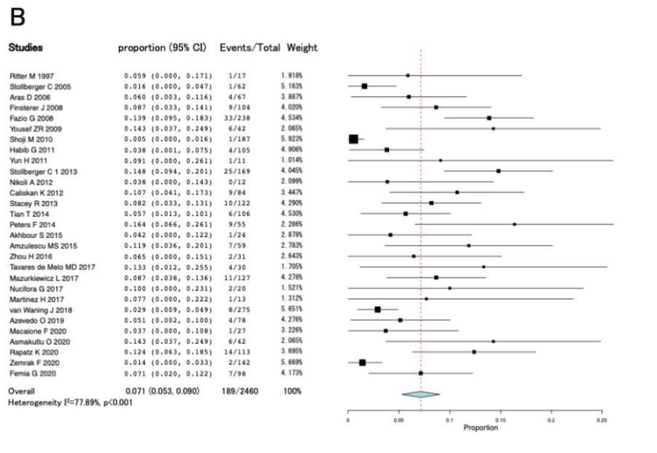
Supplemental Figure 5. Forest plot of the mean prevalence and incidence of thromboembolism in the patients with LVNC excluding studies in which antithrombotic treatment in more than 30% of patient's population

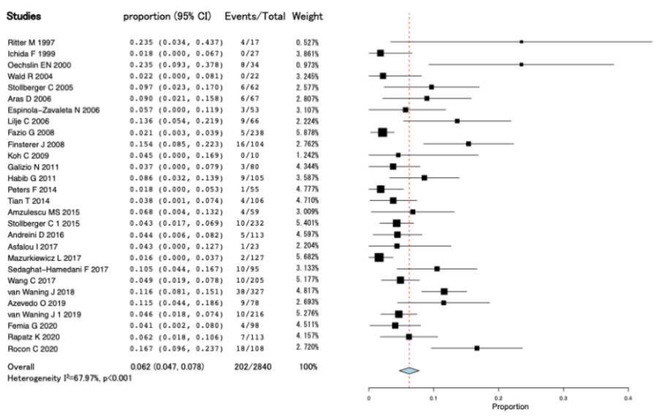
Mean prevalence of thromboembolism in all patients (A) and adult patients (B) with LVNC.

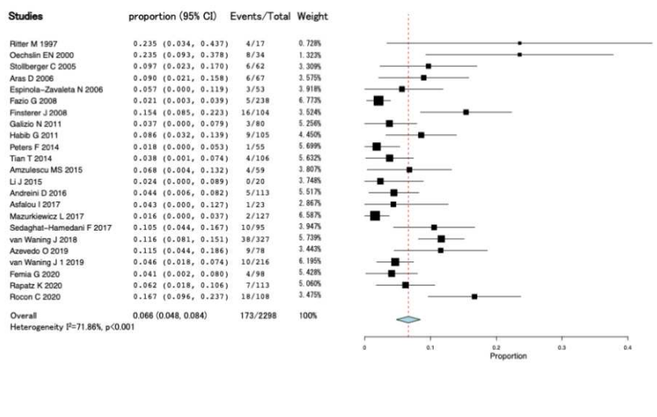
Mean incidence of thromboembolism in the whole aged (C) and adult patients (D) with LVNC.

CI, confidence interval.









Supplemental Figure 6. Forest plot of the mean mortality rates in the patients with LVNC according to subgroup analysis

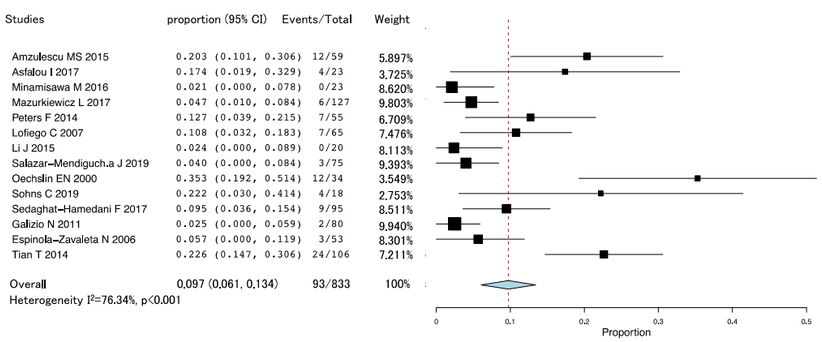
The mean mortality rates in studies with average LVEF <40% (A) and >40% (B).

The mean mortality rates in studies with mean follow-up period <4years (C) and >4years (D).

The mean mortality rates in studies with mean prevalence of AF <10% (E) and prevalence of AF >10% (F).

CI, confidence interval.

A



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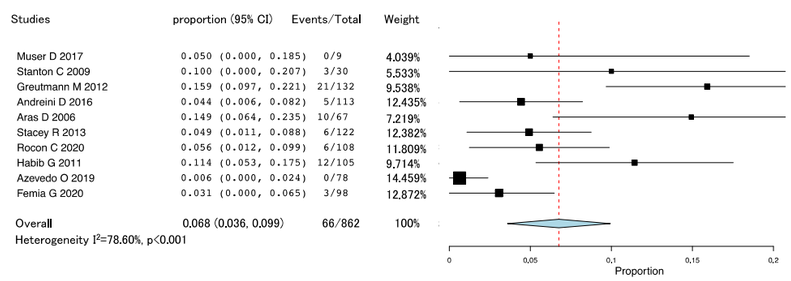

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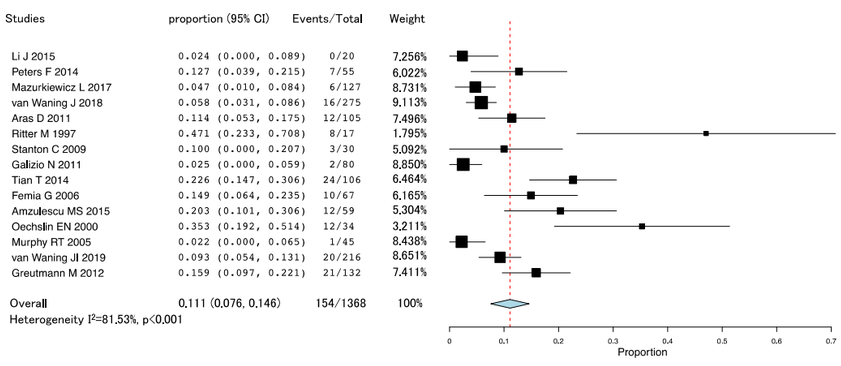

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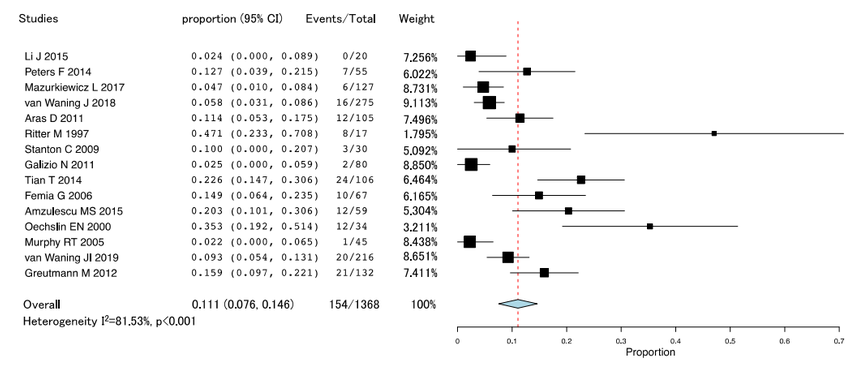

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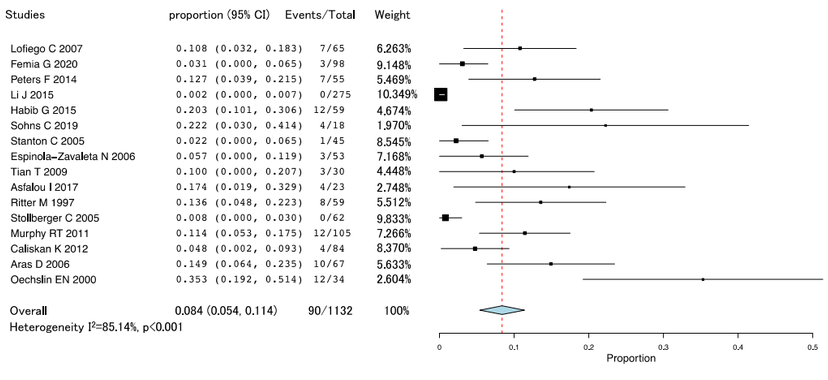

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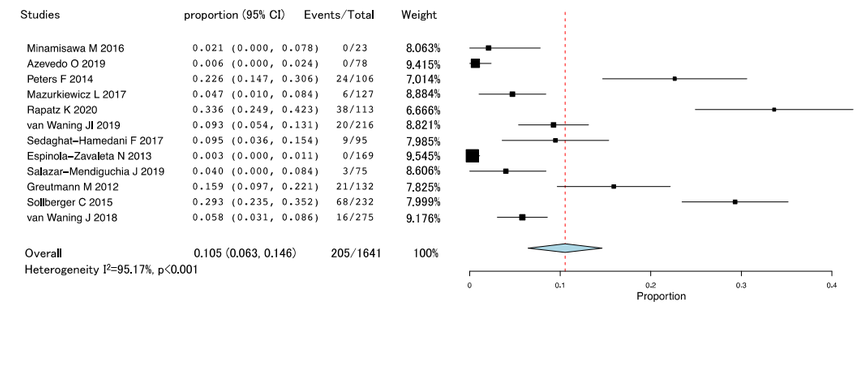


Supplemental Figure 7. Forest plot of the mean mortality and heart plantation rates in the patients with LVNC according to subgroup analysis

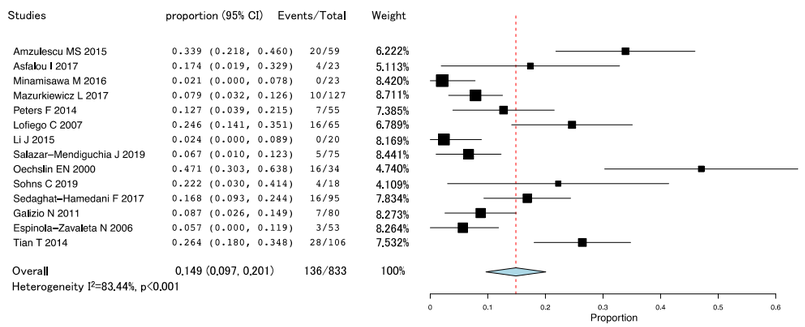
The mean mortality and heart plantation rates in studies with average LVEF <40% (A) and >40% (B).

The mean mortality and heart plantation rates in studies with mean follow-up period <4years (C) and >4years (D).

The mean mortality and heart plantation rates in studies with mean prevalence of AF <10% (E) and prevalence of AF >10% (F).

CI, confidence interval.

A



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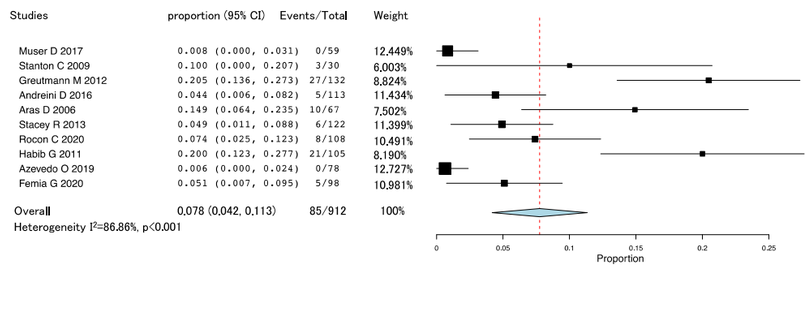

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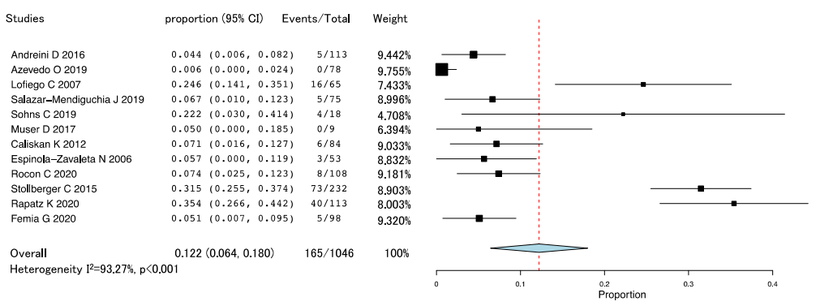

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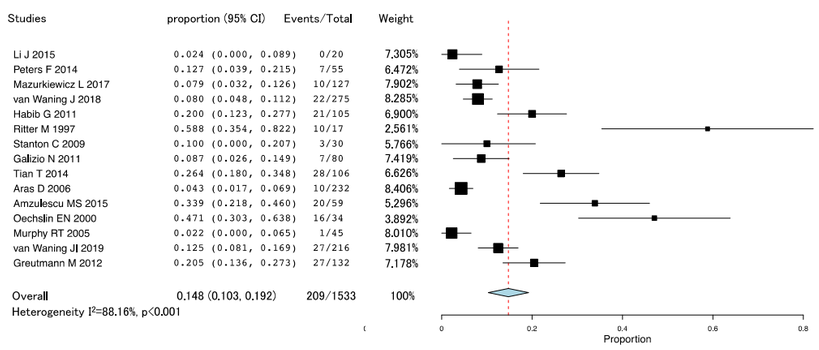

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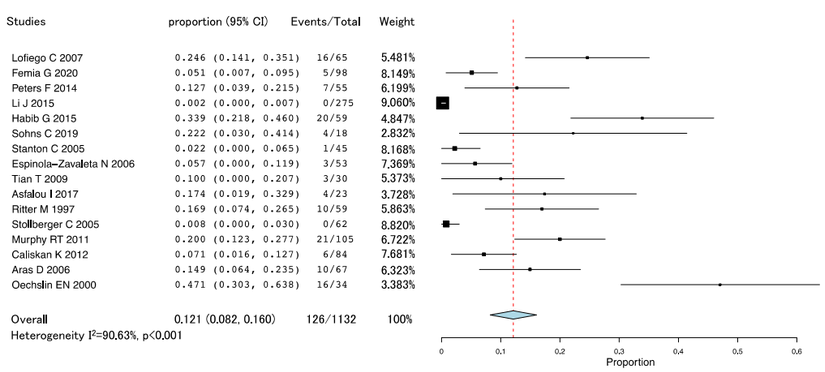

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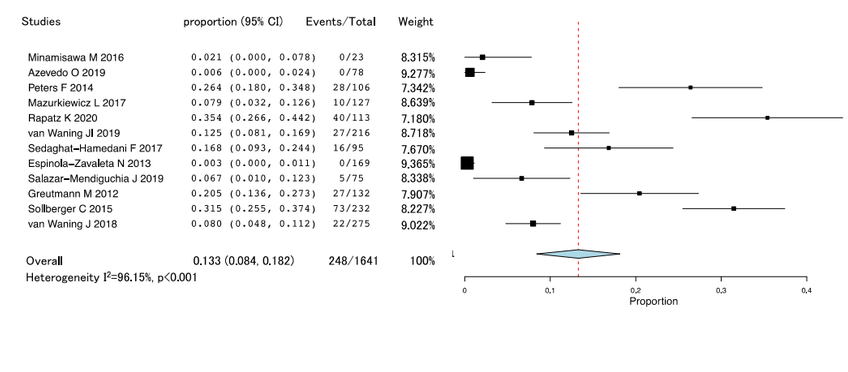

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